

Go-EVIDENCE.md — Gate Go evidence

Step 0 ("lay the rails") of the Alternate Histories Platform. This document records, for each of the seven Gate Go criteria (STEP0-PLAN.md §0), the command executed, the observed result, and pass/fail. It is the platform's first entry in its validation record. Environment: Python 3.12.10, uv 0.11.x, Windows (CI: Ubuntu). All tests run with no network (pytest-socket `--disable-socket`).

Summary: 7 / 7 criteria PASS.

Go.1 — The loop completes; replay is bit-identical

Command

```
ah world build --preset stagflation
ah world validate
ah run --paths 100
ah replay
```

Observed

- `build` → world `00000000-0000-4000-9000-000000000001`
- `validate` → `clamps=0 warnings=[] blocking=[]`
- `run` → a `run_id`; `replay` prints identical `stored` and `replay` digests → `MATCH`
- Determinism: two runs at the same seed/paths yield the same `outputs_digest`

(`tests/test_g0_end_to_end.py::test_g0_1...` , `tests/test_cli.py::test_run_two_runs_same_digest`).

Result: PASS

Go.2 — Schema + V-rule validation on every world; clamps/warnings in provenance

Command

```
uv run pytest tests/test_g0_end_to_end.py -k g0_2 -v
uv run pytest tests/test_validator.py tests/test_worldspec.py
```

Observed — `world build` stamps `provenance.validation` (`validator_version` `1.0.0` , `clamps` , `warnings`) and flips status to `validated` ; a clamp fixture records clamps via the validator (V9). Loader runs JSON Schema (Draft 2020-12) first, then pydantic; the two agree on 400 fuzzed documents.

Result: PASS

Go.3 — RunRecords store resolved engine, seed, SHA-256 digest; tamper detected

Command

```
uv run pytest tests/test_g0_end_to_end.py -k g0_3 -v
uv run pytest tests/test_stores.py -k tamper
```

Observed — RunRecord carries `resolved_engine` (`generator_version`, `validator_version` `1.0.0`, `battery_version` `battery-0.1`), `seed`, and `outputs_digest` (`sha256:...`). `verify_run` returns `True` for an intact record and `False` after the stored digest (or the stored world) is mutated.

Result: PASS

Go.4 — Chronicle is append-only (update/delete raise)

Command

```
uv run pytest tests/test_g0_end_to_end.py -k g0_4 -v
uv run pytest tests/test_stores.py -k chronicle
```

Observed — a raw `UPDATE / DELETE` on `chronicle` raises `sqlite3.Error` (trigger), and the repository module exposes no mutator functions. Both layers covered.

Result: PASS

Go.5 — Offline compiler harness: 50 fixtures, schema-valid, bounds-clamped

Command

```
uv run pytest tests/test_compiler.py
uv run pytest tests/test_g0_end_to_end.py -k g0_5 -v
```

Observed — 50 fixtures (40 valid, 5 clamp, 5 reject). Valid + clamp worlds compile, validate, build, and run ≥ 12 months with finite output; clamp worlds record clamps; reject worlds are rejected. No network (FixtureCompiler only; the live adapter is not imported).

Result: PASS

Go.6 — Validation-battery skeleton computes the panel and evaluates thresholds

Command

```
uv run python -m ah.battery.report      # CI job, stagflation preset
ah battery                             # on a stored run
```

Observed — battery battery-0.1 prints the stylized-fact panel (excess kurtosis, skew, Hill tail index, ACF $r / |r|$, max-drawdown median, corr distance) with enforce failures: 0 and exit code 0. Thresholds are all todo (non-blocking) per the plan; the enforce path is exercised in tests.

Result: PASS

Go.7 — Lint, type-check, tests $\geq 90\%$ coverage on core/, and the Go e2e test — all green

Command

```
uv run ruff check .
uv run ruff format --check .
uv run pyright
uv run pytest --cov=ah.core --cov-fail-under=90
uv run pytest tests/test_g0_end_to_end.py
```

Observed

- ruff: All checks passed!
- ruff format: 50 files already formatted
- pyright: 0 errors, 0 warnings, 0 informations
- pytest: 186 passed ; coverage on ah.core = **96.54%** (gate ≥ 90)
- Go end-to-end: 7 passed

Result: PASS

All seven criteria pass. Tagging v0.1.0-g0.